

LAB MANUAL

CSE 324

Operating Systems Lab

Course Teacher

Nushrat Jahan Oyshi

Lecturer

Computer Science & Engineering

Daffodil International University

Lab 1

- Open the terminal: ctrl + alt + T
- Search the shell type: ~\$ cat /etc/shells

Output:

/ bin / sh

/ bin / dash

/ bin / bash

/bin / rbash

So these are the shells your system will support

sh shell: bourne shell, this is the original shell that is used in the unix system or unix-like environment.

bash shell: bourne again shell which is an improved version of the sh shell, it is considered the standard shell which is intuitive and flexible.

So nowadays it's the most used shell in linux operating systems or unix based systems.

- Search bash location: ~\$ which bash

Output:

/ bin / bash

How to Add a Comment:

This is a comment

Change and Create Directory:

- ~\$ cd Desktop /
Changes the directory to Desktop
- ~/Desktop\$ mkdir oslab
Creates a new folder or directory in current location

Create Your First Bash Script:

- ~/Desktop\$ touch [hello.sh](#)
This command creates a new file and here, hello.sh is the file name
- ~/Desktop\$ touch [hello](#)
This command will also work but it is good practice to keep the file extension.
- ~/Desktop\$ ls
Shows all the existing files in current directory
- ~/Desktop\$ ls -al
Shows read and write permissions for the file
Output:
- rw - rw - r - - this means, read and write for the owner, read and write for the owner group, and only read for the user
- ~/Desktop\$ clear
clears the terminal

Editor: Visual Studio Code

- ~/Desktop\$ code .
This will open the directory using the vs code editor

The hello.sh file will open in vs code

If you don't have vs code, double click on the shell script and it will open the default editor in your system.

Script: hello.sh

#!/bin/bash	comment: bash location
echo "hello world"	comment: echo prints whatever is in ""

Now save the file, go back to the terminal and execute hello.sh

- ~/Desktop\$./hello.sh
Output:
Permission denied

This is because you have only read and write permission, so you can't execute this file.

Change the file permission:

```
~/Desktop$ chmod +x hello.sh
```

```
~/Desktop$ ls -al
```

Output:
- rwxrwxr - x

```
~/Desktop$ ./hello.sh
```

Output:
hello world